Handwriting Image-to-Text: Comparing CRAFT+TrOCR and Qwen2.5-VL on IAM Lines and Real-World Photos

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Abstract

We study handwriting recognition from images in a practical "image-to-text" setting by comparing a traditional two-stage OCR pipeline (CRAFT text detection followed by TrOCR recognition) with a modern vision-language model (Qwen2.5-VL 3B (Qwen Team 2026))prompted for transcription.

On the Teklia/IAM-line dataset (10,373 line images), TrOCR without detection or cropping achieves the best performance (CER 1.21%, WER 3.63%), while Qwen2.5-VL achieves competitive character-level accuracy (CER 5.97%) but exhibits higher word-level errors (WER 29.26%). When applying CRAFT detection to simulate real-world conditions—where TrOCR alone cannot localize text—performance degrades significantly (CER 29.27%, WER 82.40%), indicating that detection can introduce error propagation when applied to already segmented line images.

We further evaluate standalone TrOCR, CRAFT+TrOCR, and Qwen2.5-VL on real-world document images. TrOCR alone fails to produce meaningful outputs due to the lack of text localization, while CRAFT+TrOCR generates partially readable but imperfect transcriptions. In contrast, Qwen2.5-VL demonstrates more robust performance on real-world images, producing coherent transcriptions that better preserve the original meaning.

These results highlight a trade-off between modular OCR pipelines and end-to-end vision-language models, where traditional methods excel on clean, pre-segmented data, while VLMs offer greater robustness in unconstrained real-world scenarios.

Introduction

Handwriting OCR remains challenging due to diverse writing styles, imaging conditions, and ambiguous character shapes. In many course and product settings, the goal is not just character-level accuracy but reliable end-to-end transcription from a photo of handwriting into editable text.

We compare two families of solutions: (1) a classic modular OCR pipeline that first detects text regions and then recognizes each crop, versus (2) an end-to-end vision-language model (VLM) that can be prompted to directly transcribe an image.

In addition to accuracy, we consider practical deployment constraints such as data privacy. All models used in this work are executed with downloaded pre-trained weights, and inference is performed without transmitting images or text to external servers. This setup avoids potential privacy risks associated with cloud-based OCR services, making the pipeline suitable for sensitive handwritten content.

Our contributions are: (i) an evaluation of three inference-time baselines on IAM line images (Qwen2.5-VL, CRAFT+TrOCR, and TrOCR-only), (ii) an implementation that supports both "detect+crop" and "no-crop" TrOCR modes under a single interface, (iii) a Colab-ready workflow that makes full-dataset evaluation feasible on a GPU, and (iv) a privacy-preserving, fully local inference pipeline that does not require sending user data to external APIs.

Background and Related Work

CRAFT (Character Region Awareness for Text Detection). CRAFT is a scene-text detection model designed to localize individual character regions and the affinity between adjacent characters (Baek et al. 2019). It employs a convolutional neural network (CNN) backbone to generate two heatmaps: a character region score map and an affinity score map that links neighboring characters. These maps are post-processed to form word-level bounding polygons, which can then be cropped and passed to a recognition model. By operating at the character level, CRAFT is effective in handling irregular layouts, curved text, and complex backgrounds commonly found in real-world images.

TrOCR (Transformer-based OCR). TrOCR is an end-to-end text recognition model built on the Transformer architecture (Li et al. 2021). It consists of a vision encoder, typically a Vision Transformer (ViT), that converts an input image into a sequence of visual embeddings, and a text decoder that autoregressively generates the output sequence token by token. Pre-trained on large-scale synthetic and real-world text datasets, TrOCR achieves strong performance on handwriting recognition tasks. However, it assumes that the input image contains a single, well-cropped text region, which motivates pairing it with a detection model such as CRAFT for full-page or unconstrained images.

Vision-Language Models (Qwen2.5-VL). Recent vision-language models (VLMs) extend the Transformer framework to jointly process visual and textual inputs. Qwen2.5-VL follows this paradigm by combining a vision encoder

with a large language model, enabling it to accept images and generate text conditioned on natural-language prompts. Unlike modular OCR pipelines, Qwen does not explicitly separate detection and recognition; instead, it attends over the entire image and implicitly models layout, context, and reading order through pre-training. This enables direct end-to-end transcription, though it may introduce variability in structured outputs such as tables or multi-line documents.

Model Selection Rationale. This project intentionally selects widely used, reproducible, and locally executable models to enable a fair and easily replicable comparison.

For the modular OCR pipeline, we use the standard CRAFT detector ("craft_mlt_25k.pth") together with the TrOCR recognizer ("microsoft/trocr-large-handwritten"). The chosen CRAFT checkpoint is a commonly used public model for general-purpose scene text detection, providing a consistent and well-established detection baseline across experiments. The TrOCR model is a widely adopted Transformer-based handwriting recognizer available through Hugging Face, and serves as a strong representative of modern OCR recognition systems when paired with an explicit detection stage.

For the end-to-end approach, we use the vision-language model "Qwen/Qwen2.5-VL-3B-Instruct". This model is selected because it is sufficiently lightweight to run while still benefiting from large-scale multimodal pretraining and instruction tuning. Unlike the modular pipeline, Qwen2.5-VL performs direct image-to-text transcription without requiring an explicit detection stage, making it a suitable representative of end-to-end vision-language approaches.

These model choices allow us to compare two contrasting paradigms—explicit modular OCR pipelines versus implicit end-to-end transcription—under both controlled (IAM-line) and real-world settings, while ensuring that all experiments remain reproducible and privacy-preserving through fully local inference. We report exact model identifiers and checkpoints, as these choices materially affect both accuracy and runtime.

Pipelines

Traditional OCR Baseline: CRAFT + TrOCR

The baseline follows a two-stage design: (1) CRAFT detects text regions and outputs quadrilateral polygons, (2) each polygon is cropped from the original image and fed into TrOCR to decode text, and (3) recognized crop texts are concatenated in estimated reading order.

A practical issue is that datasets like IAM-line already provide single-line images; running a detector can fragment a line into multiple partial crops. To handle both line images and real-world photos, our code exposes a toggle `--use_craft`: when disabled, TrOCR runs once on the full image (no detection/cropping).

End-to-End Baseline: Qwen2.5-VL

Qwen2.5-VL is used as a direct transcription model: given an image and a prompt ("Transcribe the handwritten text exactly; output only the text"), it generates a text sequence. Unlike the modular baseline, Qwen can condition on the whole image and potentially use global context (layout, neighboring words) during decoding. For practical inference on large real-world photos, we optionally resize inputs so the maximum image dimension is capped (preserving aspect ratio), which helps reduce GPU memory usage and improves throughput.

Evaluation Setup

Dataset

We evaluate on the Teklia/IAM-line dataset, a standard benchmark for handwritten text recognition consisting of segmented line images (Teklia 2026). The dataset contains 10,373 line images of English handwriting with corresponding ground-truth transcriptions. We concatenate all available splits to form a large-scale evaluation set without any fine-tuning, focusing purely on inference-time performance.

To complement this controlled benchmark, we also collect a small set of real-world handwritten document images. These images include handwritten text on paper, notebooks, and whiteboards captured using mobile phone cameras. Compared to IAM-line, these images exhibit more challenging conditions such as multiple lines of text, varying lighting, perspective distortion, background clutter, and inconsistent handwriting styles. Unlike the IAM dataset, these samples are not pre-segmented into individual lines, making them a useful qualitative testbed for evaluating robustness in practical scenarios.

Metrics

We evaluate transcription quality using Character Error Rate (CER) and Word Error Rate (WER), computed with the `jiwer` library (jitsi/jiwer contributors 2026). Both metrics measure the edit distance between the predicted transcription and the ground truth, normalized by the length of the reference text.

Character Error Rate (CER) is defined as the minimum number of character-level insertions, deletions, and substitutions required to transform the predicted text into the ground truth, divided by the total number of characters in the reference. CER captures fine-grained transcription accuracy and is more tolerant of small mistakes.

Word Error Rate (WER) applies the same edit-distance formulation at the word level. Because errors are counted per word, WER is a stricter metric: even a minor character error can cause an entire word to be considered incorrect. As a result, WER better reflects the readability and usability of the transcription in downstream applications.

Lower values for both CER and WER indicate better performance. We report both metrics to capture complementary aspects of model behavior: CER reflects local character-level accuracy, while WER highlights the impact of errors on semantic correctness.

Implementation and Reproducibility

Our repository provides: (1) scripts to run each pipeline on a folder or single image, (2) evaluation scripts that emit per-sample CSVs and summary text, and (3) Colab notebooks

Table 1: IAM-line results (10,373 samples). Lower is better.

Method	CER (%)	WER (%)
TrOCR-only (no detection/cropping)	1.21	3.63
Qwen2.5-VL-3B-Instruct	5.97	29.26
CRAFT + TrOCR (detect+crop)	29.27	82.40

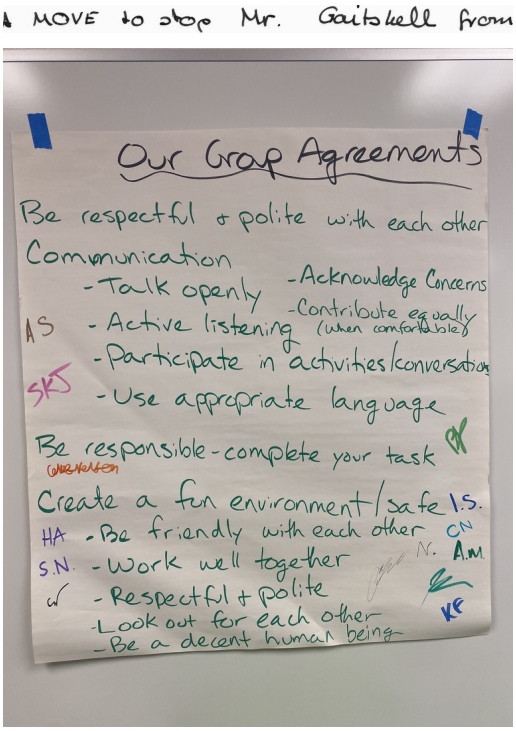

Figure 1: Top: IAM-line example (clean, single-line input). Bottom: real-world handwritten image with multiple lines, distortion, and noise, highlighting the domain gap between benchmark datasets and practical inputs.

that install dependencies, cache Hugging Face weights, and run evaluations on GPU.

Key implementation details include: (i) defensive cropping for out-of-bounds polygons, (ii) deterministic reading-order sorting of detected boxes, and (iii) optional disabling of CRAFT for line images.

Results

Table 1 summarizes results on IAM-line.

On IAM-line, the simplest traditional baseline—TrOCR applied directly to the full line image—performs best. This is expected: IAM-line images are already cropped to a single text line, closely matching the input distribution used by handwritten TrOCR models.

As shown in Figure 1, real-world images exhibit significantly more variability than IAM-line, including multi-line structure and environmental noise.

In contrast, adding CRAFT detection on IAM-line substantially degrades accuracy. A likely explanation is that detector-driven cropping fragments a single line into multiple partial crops, introduces ordering errors, and can cut through characters, all of which harm downstream recognition.

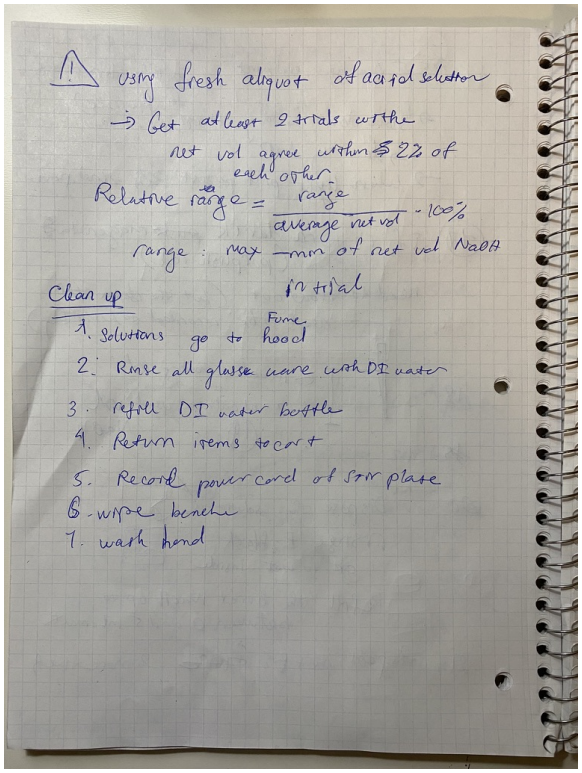

Figure 2: Real-world handwritten image used for qualitative comparison of model predictions.

This highlights a key mismatch between model assumptions and input distribution: TrOCR is optimized for pre-segmented line images, while real-world inputs violate this assumption.

Qwen2.5-VL achieves strong character-level accuracy but a higher word error rate than TrOCR-only. This suggests many predictions are close (small character substitutions or spacing differences) yet still counted as word-level mistakes.

While TrOCR-only is ideal for already-segmented line images, it often underperforms on real-world photos of handwriting (e.g., a phone photo of a page) because it lacks a built-in text detection stage. In these images, text may be multi-line, curved, unevenly illuminated, partially occluded, or embedded in background clutter (Figure 1). Feeding the entire page to a recognizer trained on line-like crops forces it to implicitly solve detection, layout, and reading order, which can lead to short, nonsensical outputs.

Figure 2 shows a real-world handwritten image with multiple lines, irregular spacing, and background noise. We qualitatively compare the predictions produced by the two models on this same input.

CRAFT+TrOCR Output on Figure 2: “using fresh . alrg . not safacr . ful seduation . 6et . at least 2 . irrals . cuthe must vol . agree . urthim . s 2/ of each ’ other . relative . range . is range . joo average . net . vol . range . may imm

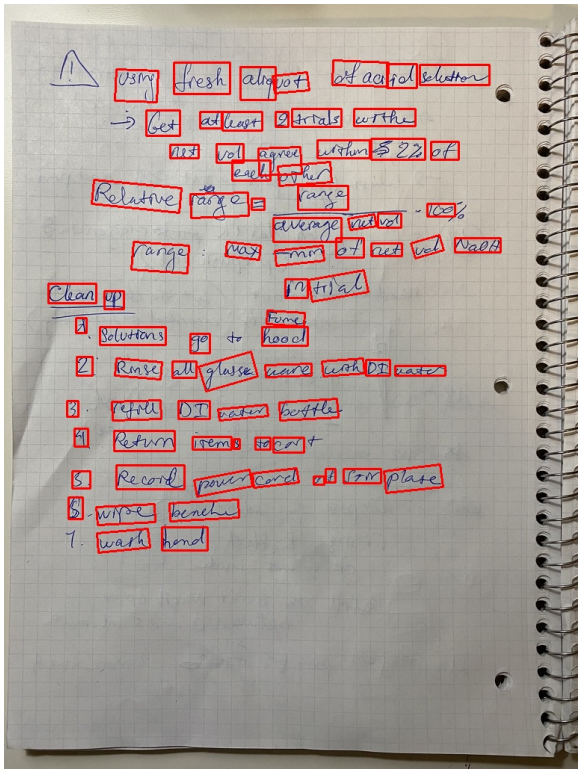

Figure 3: Output of Craft before being crop and reorder and individually feed into TrOCR as input.

. of net . vet . nalt clean . up frv . total . af fume . solutions . go . hood . 2 . rumse . all . glusse name . catch . di . water 3 . refru . dt . water bottle . 4) . return . 1perm . g . ito . cor s . record . power . carel . or ff . strir . plate . ts . whose . benehe wash shand ."

Qwen2.5-VL Output on Figure 2: "A) using fresh aliquot of acqol solution → Get at least 2 trials with the net vol agree within ± 2% of each other. Relative range = range average net vol - 100% range : max – min of net vol NaOH Clean up 1. Solutions go to hood 2. Rinse all glassware with DI water 3. refill DI water bottle 4. Return items to cart 5. Record power cord of fan plate 6. wipe bench 7. wash hand"

These examples highlight that CRAFT+TrOCR produces highly fragmented and noisy transcriptions under real-world conditions, likely due to error propagation from imperfect detection and cropping. Additionally, processing each detected region independently removes global context, further degrading recognition quality and consistency (Figure 3).

In contrast, Qwen2.5-VL produces more coherent and structured outputs, better preserving the overall semantic content despite minor inaccuracies.

Limitations and Future Work

This project evaluates off-the-shelf models at inference time without task-specific fine-tuning. As a result, the reported numbers primarily reflect dataset–model fit and pipeline design choices rather than the best achievable performance with adaptation. In addition, while the system is intended to run locally, initial model downloads typically require internet access; our experiments also use Colab for GPU convenience, which differs from fully offline deployment.

Our quantitative evaluation is limited to IAM-line, which consists of already-segmented line images. This setting strongly favors recognizers such as TrOCR when run directly on the full line image, and it does not fully capture real-world challenges (multi-line layouts, perspective distortion, clutter, and unknown reading order). For real-world photos, we therefore include only qualitative examples in this paper, since a curated benchmark with ground-truth transcriptions is not yet available.

Future work should: (1) build a small real-world handwriting photo benchmark with ground-truth transcriptions to enable CER/WER measurement and fair comparison across methods, (2) improve the modular OCR pipeline for photos by tuning CRAFT thresholds, using more robust grouping and reading-order logic, and recognizing at the line level when possible, (3) run controlled ablations on prompting and decoding for Qwen2.5-VL to quantify sensitivity and improve reliability, and (4) explore lightweight post-recognition correction and formatting (e.g., normalization and structured document export) while reporting both raw and post-processed outputs.

Conclusion

We compare a modular OCR pipeline (CRAFT+TrOCR), a recognition-only baseline (TrOCR), and an end-to-end vision-language model (Qwen2.5-VL) for handwriting transcription.

On IAM-line, TrOCR achieves the best performance (CER 1.21%, WER 3.63%), while adding CRAFT degrades accuracy due to error propagation. Qwen2.5-VL shows higher WER but remains more robust on real-world images, producing more coherent and semantically meaningful outputs.

These results highlight a trade-off between modular OCR systems, which excel on clean inputs, and end-to-end VLMs, which are more flexible in unconstrained settings. Future work includes improving layout-aware recognition and evaluating on curated real-world benchmarks.

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